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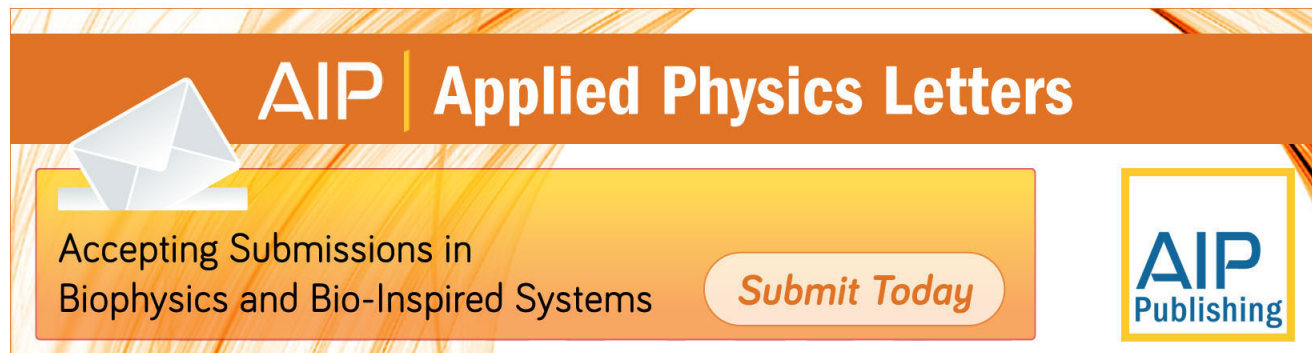
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Approximate Expression for the Tunneling Correction*

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By using an approximate method, the tunneling-correction factor for the unsymmetrical Eckart potential function and its temperature dependence are formulated in forms which are capable of simple numerical calculations. The method is designed to evaluate the integral which includes a violently changing exponent. The result is compared with Johnston's results obtained by electronic computation; the agreement is good in chemically interesting cases.

IN view of recent interest¹⁻³ in the tunneling correction of reaction rate formulated using activated-complex theory, it seems worth pointing out that we can formulate a useful expression of the correction for unsymmetrical Eckart potential function⁴ in a form which can be simply evaluated. Using the potential

$$W(x) = -AY/(1-Y) - BY/(1-Y)^2$$

$$Y = -\exp(2\pi x/l),$$

where x is the Cartesian coordinate, l is a characteristic length, and A and B are constants, Eckart⁴ found the probability $\mathcal{K}(E)$ that a particle starting toward the barrier with energy E at $x = -\infty$ will pass the barrier and appear later at $x = +\infty$ with energy E . This was found by solving the wave equation

$$\mathcal{K}(E) = 1 - \frac{\cosh[2\pi(\alpha - \beta)] + \cosh 2\pi\delta}{\cosh[2\pi(\alpha + \beta)] + \cosh 2\pi\delta}, \quad (1)$$

where

$$\alpha = \frac{1}{2}(E/C)^{1/2}, \quad \beta = \frac{1}{2}[(E-A)/C]^{1/2},$$

$$\delta = \frac{1}{2}[(B-C)/C]^{1/2}, \quad C = \hbar^2/8ml^2,$$

and m is the mass of the incident particle.

We consider the case in which l is large and C is very small. Then,

$$\cosh[2\pi(\alpha + \beta)] \simeq \frac{1}{2} \exp[2\pi(\alpha + \beta)],$$

$$\sinh(2\pi\alpha) \simeq \frac{1}{2} \exp(2\pi\alpha),$$

$$\cosh(2\pi\delta) \simeq \frac{1}{2} \exp[\pi(B/C)^{1/2}], \quad \sinh(2\pi\beta) \simeq \frac{1}{2} \exp(2\pi\beta).$$

Hence, the probability becomes

$$\mathcal{K}(E) = \frac{\exp[2\pi(\alpha + \beta) - \pi(B/C)^{1/2}]}{\exp[2\pi(\alpha + \beta) - \pi(B/C)^{1/2}] - 1}.$$

With a Boltzmann distribution of incident particles at

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¹ R. P. Bell, Trans. Faraday Soc. **55**, 1 (1959).

² H. S. Johnston and D. Rapp, J. Am. Chem. Soc. **83**, 1 (1961).

³ H. S. Johnston and J. Heicklen, J. Phys. Chem. **66**, 532 (1962).

⁴ C. Eckart, Phys. Rev. **35**, 1303 (1930).

$x = -\infty$, the tunneling correction Γ can be expressed in an integrated form over all possible energy of the incident particle⁵;

$$\Gamma = \exp\left(\frac{\Delta W_1}{kT}\right) \int_0^\infty \mathcal{K}(E) \exp\left(\frac{-E}{kT}\right) d\left(\frac{E}{kT}\right),$$

where ΔW_1 is the maximum value of $W(x)$ measured from the value at $x = -\infty$. Then the integral to be evaluated is

$$I = \int_0^\infty \exp[f(E)] dE,$$

with

$$f(E) = 2\pi(\alpha + \beta)$$

$$- \log\{\exp[2\pi(\alpha + \beta) - \pi(B/C)^{1/2}] + 1\} - E/kT.$$

The exponent in the integral has a maximum at $E = E^*$. Unless $f(E)$ is a constant, at least one of the higher derivatives $f''(E^*)$, $f'''(E^*)$, $\dots$ differs from zero. Let k be the smallest positive integer such that $f^{(k)}(E^*) \neq 0$. We shall assume that $k = 2$. The cases $k > 2$ are only slightly more difficult to treat, and as they hardly occur in applications we shall disregard them. The situation in a small neighborhood of E^* is then mainly determined by the value $f''(E^*)$, since

$$f(E) = f(E^*) + \frac{1}{2}f''(E^*)(E - E^*)^2 + \dots$$

When we substitute this expression in the integral, we have the approximate expression for $f''(E^*) < 0$:

$$I = [2\pi/f''(E^*)]^{1/2} \exp[f(E^*)],$$

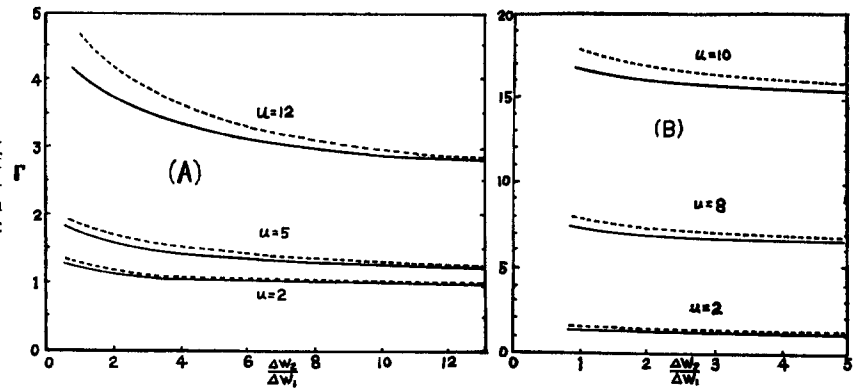
where E^* is the solution of

$$E^* \frac{1}{2} [g(E^*) - \frac{1}{2}(B/E^*)^{1/2}]$$

$$+ (C^{1/2}/2\pi) \log\left\{\frac{(CE^*)^{1/2}[1 - (A/4E^*)]}{\pi kT}\right\} = 0,$$

with $g(E^*) = 1 - (A/4E^*) - (A/4E^*)^2$. Simplifying

FIG. 1. Calculated values of the tunneling correction as a function of $\Delta W_2/\Delta W_1$. The solid lines give the author's result; the dotted lines are taken from Ref. 3. (A): $2\pi\Delta W_1/h\nu=1$; (B): $2\pi\Delta W_1/h\nu=4$.



$f(E^*)$ and $f''(E^*)$ in terms of this expression, we obtain where E^* is

$$\Gamma = (2\pi)^{\frac{1}{2}} \left\{ 1 + \frac{kT}{2E^*} \left[1 + \frac{1}{2} \left(\frac{A}{E^*} \right) + \frac{5}{8} \left(\frac{A}{E^*} \right)^2 \right] - \frac{\pi kT}{(CE^*)^{\frac{1}{2}}} \left[1 + \frac{1}{4} \left(\frac{A}{E^*} \right) + \frac{3}{16} \left(\frac{A}{E^*} \right)^2 \right] \right\}^{-1} \times \frac{\exp[2\pi(E^*/C)g(E^*) + (\Delta W_1/kT) - (E^*/kT)]}{\exp[2\pi(E^*/C)g(E^*)] + \exp[\pi(B/C)^{\frac{1}{2}}]}. \quad (2)$$

A less accurate but useful approximate expression can be formulated when $\exp[2\pi(\alpha+\beta) - \pi(B/C)^{\frac{1}{2}}] + 1$ slowly changes compared with $\exp[2\pi(\alpha+\beta)]$. In this case $\exp[2\pi(\alpha+\beta) - \pi(B/C)^{\frac{1}{2}}] + 1$ has insignificant effect in controlling the over-all magnitude of the integral so that this slowly varying coefficient can be replaced by its value at $E=E^*$ at the beginning of the calculation. Therefore, $f(E)$ now is

$$(\pi/C)^{\frac{1}{2}} [E^{\frac{1}{2}} + (E-A)^{\frac{1}{2}}] - E/kT,$$

and the result is that

$$\Gamma' = 2 \left(\frac{C}{E^*} \right)^{\frac{1}{2}} \left(\frac{E^*}{kT} \right) \left[1 - \frac{3}{8} \left(\frac{A}{E^*} \right) - \frac{3}{16} \left(\frac{A}{E^*} \right)^2 \right] \times \frac{\exp[2\pi(E^*/C)g(E^*) + (\Delta W_1/kT) - (E^*/kT)]}{\exp[2\pi(E^*/C)g(E^*)] + \exp[\pi(B/C)^{\frac{1}{2}}]}. \quad (3)$$

$$\theta_r = \frac{E^* - \Delta W_1}{kT} - \frac{(kT/2E^*) \left[1 + \frac{1}{2} (A/E^*) + \frac{5}{8} (A/E^*)^2 \right] - [\pi kT/(CE^*)^{\frac{1}{2}}] \left[1 + \frac{1}{4} (A/E^*) + \frac{3}{16} (A/E^*)^2 \right]}{2 \left\{ 1 + (kT/2E^*) \left[1 + \frac{1}{2} (A/E^*) + \frac{5}{8} (A/E^*)^2 \right] - [\pi kT/(CE^*)^{\frac{1}{2}}] \left[1 + \frac{1}{4} (A/E^*) + \frac{3}{16} (A/E^*)^2 \right] \right\}}. \quad (4)$$

Similarly, for Eq. (3),

$$\theta_r = (E^* - \Delta W_1 - kT)/kT. \quad (5)$$

With the analytical expression for E^* given above, Eq. (5) becomes

$$\theta_r = \left(\frac{\pi^2 kT}{C} \right) \left[1 + \frac{AC}{2(\pi kT)^2} + \frac{3}{16} \frac{(AC)^2}{(\pi kT)^4} \right] - \frac{\Delta W_1}{kT} - 1. \quad (6)$$

$$[(\pi kT)^2/C] [1 + AC/2(\pi kT)^2 + 3(AC)^2/16(\pi kT)^4].$$

It can be seen that the exponential parts of Eqs. (2) and (3) are identical. In general, the replacement of slowly changing E -dependent quantities by their values at $E=E^*$ at the beginning of the integration leads only to some changes in the pre-exponential part. Equation (3) gives correct temperature dependency of Γ as in Eq. (2), and it is this equation which is appropriate for most of the numerical calculations.

The theoretical expressions for a rate constant have the form

$$k = B(T) \exp(-W/kT).$$

Since the empirical Arrhenius activation energy is

$$E_a = -R \frac{d \log k}{d(1/T)},$$

we have $E_a = W + \theta kT$, where θ is $d \log B/d \log T$. In activated-complex theory θ is additive; that is,

$$\theta = \theta_{tr} + \theta_{rot} + \theta_{vib} + \theta_r + \dots$$

In this expression $\theta_r = d \log \Gamma/d \log T$ and it can be explicitly determined using Eq. (2) or Eq. (3). For Eq. (2) we obtain

If we write the rate constant as $k = \Gamma A^* \exp(-W/kT)$, for Eq. (3),

$$A_a/A^* = \Gamma' \exp(\theta_r), \quad (7)$$

where A_a is the apparent pre-exponential factor in the Arrhenius expression. Substituting Eqs. (3) and (5)

TABLE I. Calculated values of Γ' as a function of $\Delta W_1/kT$.

$\Delta W_1/kT$	0.2	0.4	0.6	0.8	1.0	1.4	2.0
(a) ^a	1.08	1.32	1.61	1.97	2.94	4.45	8.01
(b)	1.07	1.31	1.59	1.95	2.38	3.57	6.48
(c)	0.86	1.15	1.40	1.71	2.09	3.13	5.69

^a (a): $E^*/kT=9$, $A/E^*=0.2$, $B/C=5$, $C/E^*=0.05$,(b): $E^*/kT=12$, $A/E^*=0.2$, $B/C=10$, $C/E^*=0.01$,(c): $E^*/kT=12$, $A/E^*=0.5$, $B/C=10$, $C/E^*=0.01$.

into Eq. (7) we then obtain

$$\frac{A_a}{A^*} = 2 \left(\frac{C}{E^*} \right)^{\frac{1}{2}} \left(\frac{E^*}{kT} \right) \left[1 - \frac{3}{8} \left(\frac{A}{E^*} \right) - \frac{3}{16} \left(\frac{A}{E^*} \right)^2 \right] \\ \times \frac{\exp[2\pi(E^*/C)g(E^*) - 1]}{\exp[2\pi(E^*/C)g(E^*)] + \exp[\pi(B/C)^{\frac{1}{2}}]}. \quad (8)$$

When the values of A , B , and C are given, all terms in Eqs. (3), (6), and (8) can be readily calculated without using sophisticated computing devices or lengthy graphical integrations. Obviously the evaluation of Eq. (2) is not simple, since we must carry out a tedious

numerical solution to obtain E^* . However, as shown below, it is sufficient to use the approximated solution, Eq. (3), instead of Eq. (2) for chemically interesting cases.

For chemically interesting values of A/E^* , C/E^* , and C/B , the calculated values of Γ' are listed in Table I as a function of $\Delta W_1/kT$. When ΔW_1 is large, only a small fraction of the reacting molecules surmounts the potential barrier, but many of them leak through the barrier so that large corrections are necessary. Table I also shows that larger corrections are necessary for smaller values of A/E^* .

It is instructive to compare the numerical results obtained from this method with those obtained from other methods. Exact values of Γ obtained by electronic-computer technique can be found in Ref. 3. The calculated values are plotted in Fig. 1 along with the exact values of Ref. 3 as a function of $\Delta W_2/\Delta W_1$ for various values of $u = h\nu/kT$. The agreement with the exact data is good except at large u values. When u is large the integral $\mathbf{I}$ has a maximum with a small curvature so that major contribution to $\mathbf{I}$ comes from a rather wide range of E . In such cases it is important to include the contribution of the third and higher derivatives in the expansion of $f(E)$ around $E = E^*$.